Chapter 1
Matrices and Systems of Equations

1.3 Matrix Arithmetic 矩阵运算
1.4 Matrix Algebra 矩阵代数

Objectives

- Matrix arithmetic
  - =, +, -, scalar multiplication
  - Matrix-vector multiplication, matrix-matrix multiplication
- Matrix algebra
  - Algebraic properties of matrix operations
  - Transpose, symmetry
  - Inverse, singular/nonsingular matrix

Notation

- Class and lectures (default by most books)
  - Lower bold face letter as column vector, $\boldsymbol{x} \in R^{n\times 1}$, and its transpose $\boldsymbol{x}^{\boldsymbol{T}}$ as a row vector.

- Textbook
  - Lower bold face letter as a column vector:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

  - Bold face with arrow as a row vector:

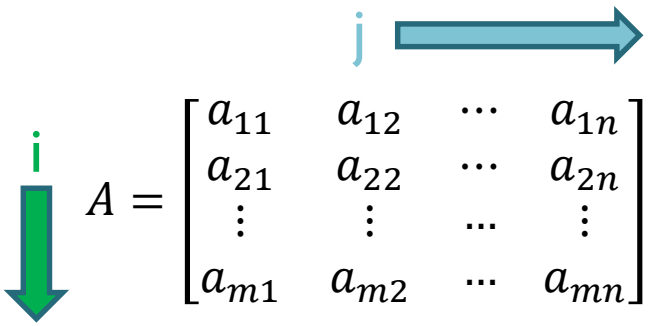
$$\vec{\boldsymbol{y}} = [y_1, y_2, \cdots y_n]$$

Notation 2

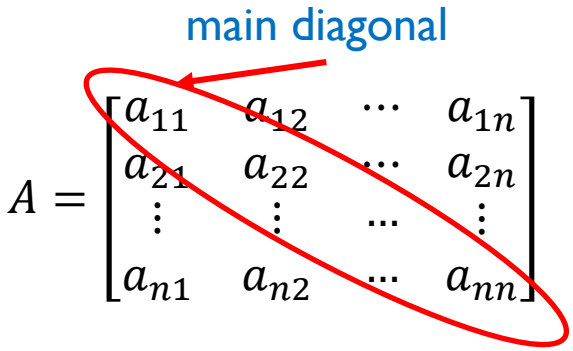
- $\boldsymbol{e}_j = [0, \dots, 0, 1, 0, \dots, 0]^T$ is the column vector whose j-th entry is 1, and other entries are 0.
- For a matrix A,
  - $A = [\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_n]$, $\boldsymbol{a}_j$ is the j-th column.
  - $A = \begin{bmatrix} \vec{\boldsymbol{a}}_1 \\ \vec{\boldsymbol{a}}_2 \\ \vdots \\ \vec{\boldsymbol{a}}_m \end{bmatrix}$, $\vec{\boldsymbol{a}}_i$ is the i-th row.

Matrix arithmetic: Equality

a_{ij}: the entry of matrix A in row i and column j.



m× n matrix A



$n \times n$ square matrix A

Definition

Two matrices are **equal** if they are of the same size and if their corresponding entries are all equal.

Thus, $A = B$ if $a_{ij} = b_{ij} \quad \forall i, j.$ (∀ denotes for every, for all)

Example

If $\begin{bmatrix} a+2b & 2a-b \\ 2c+d & c-2d \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 4 & -3 \end{bmatrix}$, find a, b, c, d.

From the matrix, we have:

a+2b = 4 , 2a-b = -2, 2c+d = 4 and c-2d = -3

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 4 \\ 2 & -1 & 0 & 0 & -2 \\ 0 & 0 & 2 & 1 & 4 \\ 0 & 0 & 1 & -2 & -3 \end{array}\right] \xrightarrow[R3 \leftrightarrow R4]{R2-2R1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 4 \\ 0 & -5 & 0 & 0 & -10 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 2 & 1 & 4 \end{array}\right] \xrightarrow[R4-2R3]{R2/-5} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 5 & 10 \end{array}\right]$$

$$\xrightarrow{R4/5} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow[R3+2R4]{R1-2R2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right]$$

R.R.E. F. (Reduced Row Echelon Form)

R.E. F. (Row Echelon Form)

So we have: a = 0, b = 2, c = 1, d = 2

Matrices arithmetic: Addition

Definition

Let A and B be matrices of the same size.

Their **sum** $A + B$ is the matrix obtained by adding together the corresponding entries of A and B.

The matrix $A + B$ will be of the same size as A and B.

If A and B are not of the **same size**, they cannot be added, and we say that **the sum does not exist**.

Thus if $C = A + B$, then $c_{ij} = a_{ij} + b_{ij} \quad \forall i, j.$

Example

Let $\mathbf{A} = \begin{bmatrix} 1 & 4 & 7 \\ 0 & -2 & 3 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 2 & 5 & -6 \\ -3 & 1 & 8 \end{bmatrix}$, and $\mathbf{C} = \begin{bmatrix} -5 & 4 \\ 2 & 7 \end{bmatrix}$.

Determine $A + B$ and $A + C$, if the sum exist.

Scalar multiplication of matrices

Definition

Let A be a matrix and c be a scalar. The **scalar multiple** of A by c, denoted cA, is the matrix obtained by multiplying every entry of A by c. The matrix cA will be the same size as A.

Example

$$\text{Let } A = \begin{bmatrix} 1 & -2 & 4 \\ 7 & -3 & 0 \end{bmatrix}.$$

$$3A =$$

Negation and Subtraction

Definition

- The matrix (-1)*C* is written –*C* and is called the **negative** of *C*.
- We define **subtraction** in terms of addition and scalar multiplication.

$$A - B = A + (-1)B$$

Example

$$C = \begin{bmatrix} 1 & 0 & -7 \\ -3 & 6 & 2 \end{bmatrix}, \text{ then } -C = \begin{bmatrix} -1 & 0 & 7 \\ 3 & -6 & -2 \end{bmatrix}$$

$$\text{Suppose } A = \begin{bmatrix} 5 & 0 & -2 \\ 3 & 6 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 8 & -1 \\ 0 & 4 & 6 \end{bmatrix}.$$

$$A - B = \begin{bmatrix} 5-2 & 0-8 & -2-(-1) \\ 3-0 & 6-4 & -5-6 \end{bmatrix} = \begin{bmatrix} 3 & -8 & -1 \\ 3 & 2 & -11 \end{bmatrix}.$$

Matrix multiplication

Given an mxn matrix A and an n-vector x, we define their product Ax (as an m-vector) so that the system of linear equation can be written as Ax = b.

In the following slides we will see this from simple case to general case:

- Row vector times column vector → scalar
- Matrix times column vector → column vector
- Matrix times matrix → matrix

Product between row and column vectors

(sometimes called the dot product, or inner product, of 2 vectors)

Given a row vector and a column vector,

$$\mathbf{a}^T = [a_1 \;\; a_2 \;\; \cdots \;\; a_n\,], \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix},$$

We define their product as a number,

$$\mathbf{a^T}\boldsymbol{x} = a_1x_1 + a_2x_2 + \cdots \;\; a_nx_n = \sum_{i=1}^{n} a_ix_i$$

Example 1: $\mathbf{a^T b} = 1(4) + 2(-1) = 2$

- If $\mathbf{a}^T = [1 \;\; 2]$, $\mathbf{b} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$, find $\mathbf{a}^T\mathbf{b}$.
- If $\mathbf{x^T} = [5 \;\; 1 \;\; 4]$, $\mathbf{y} = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}$, find $\mathbf{x}^T\mathbf{y}$.

  $\mathbf{x^T y} = \mathbf{5(3)} + \mathbf{1(4)}+4(2) = 27$

  - Meaning: total price for 3 items.



Example 2

- The linear equation

$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$
can be written as

$$[a_1, a_2, \ldots, a_n]\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = b$$

$$A\boldsymbol{x} = b$$

Where $A = [a_1, a_2, \ldots, a_n]$ is a $1 \times n$ matrix.

Example 3

- Suppose

$$e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Then

$$\boldsymbol{e}_3{}^T \boldsymbol{e}_2 = 0,$$
$$\boldsymbol{e}_2^T \boldsymbol{e}_2 = 1 = \boldsymbol{e}_3^T \boldsymbol{e}_3$$

In general, the product

$$\boldsymbol{e}_i^T \boldsymbol{e}_j = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Example before matrix-vector multiplication

- Knowing the price and volume of each drink, and knowing how many drinks there are, calculate the total price and total volume.

	coconut water	cane juice	milk			quantity
price (¥)	5	1	4		coconut water	3
volume (ml)	330	250	200		cane juice	4
					milk	2

- Total price is the product:

$$\begin{bmatrix}5 & 1 & 4\end{bmatrix}\begin{bmatrix}3\\4\\2\end{bmatrix}=5\times3+1\times4+4\times2=27$$

- Total volume is also a product:

$$\begin{bmatrix}330 & 250 & 200\end{bmatrix}\begin{bmatrix}3\\4\\2\end{bmatrix}=330\times3+250\times4+200\times2=2390$$

- Matrix-vector multiplication is doing 2 products at the same time:

$$\begin{bmatrix}5 & 1 & 4\\330 & 250 & 200\end{bmatrix}\begin{bmatrix}3\\4\\2\end{bmatrix}=\begin{bmatrix}27\\2390\end{bmatrix}$$

price

volume

Matrix-vector multiplication

If A is the m x n matrix and x is the n-vector

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

Then we define the matrix-vector product A**x** as the m-vector

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}$$

In other words, if $\vec{a}_i$ is the i-th row of A, then

$$A\boldsymbol{x} = \begin{bmatrix} \vec{a}_1\boldsymbol{x} \\ \vec{a}_2\boldsymbol{x} \\ \vdots \\ \vec{a}_m\boldsymbol{x} \end{bmatrix}$$

Also, if $\boldsymbol{a}_j$ is the j-th column of A, then

$$A\boldsymbol{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\boldsymbol{a}_n$$

Linear system can be represented as a matrix-vector equation

$$A\mathbf{x} = \mathbf{b}$$

E.g,

$$\begin{aligned} 2x_1 - 3x_2 - 4x_3 &= 15 \\ 2x_1 + 4x_2 + 6x_3 &= -12 \\ 3x_1 + 4x_2 + 5x_3 &= -8 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -3 & -4 \\ 2 & 4 & 6 \\ 3 & 4 & 5 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \mathrm{b} = \begin{bmatrix} 15 \\ -12 \\ -8 \end{bmatrix}$$

As a matrix equation: $\begin{bmatrix} 2 & -3 & -4 \\ 2 & 4 & 6 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 15 \\ -12 \\ -8 \end{bmatrix}$

As a vector equation: $x_1 \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 4 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ -12 \\ -8 \end{bmatrix}$

From matrix-vector to matrix-matrix multiplication

- If two linear systems share the same coefficient matrix but with different right-hand sides, say,

$$A\boldsymbol{x} = \boldsymbol{b}_1 \text{ and } A\boldsymbol{y} = \boldsymbol{b}_2,$$

then we can solve the two linear systems at the same time, by writing the augmented matrix together:

$$[A \mid \boldsymbol{b}_1\ \boldsymbol{b}_2]$$

- In matrix equation form:

$$A[\boldsymbol{x}\ \boldsymbol{y}] = [\boldsymbol{b}_1\ \boldsymbol{b}_2]$$

  where A, $[\boldsymbol{x}\ \boldsymbol{y}]$, $[\boldsymbol{b}_1\ \boldsymbol{b}_2]$ are all matrices.
- So, doing several matrix-vector multiplications in parallel can be rewritten as one matrix-matrix multiplication (see next slide for a definition)
- Dot product $\xrightarrow{\text{in parallel}}$ matrix-vector multiplication $\xrightarrow{\text{in parallel}}$ matrix multiplication

Example: 鸡兔同笼

	鸡	兔
头	1	1
脚	2	4

A

	小笼	大笼
鸡	2	5
兔	3	4

x_1 x_2

	小笼	大笼
头	?	?
脚	?	?

b_1 b_2

$$\begin{bmatrix}1 & 1\\2 & 4\end{bmatrix}\begin{bmatrix}2 & 5\\3 & 4\end{bmatrix}=\begin{bmatrix}5 & 9\\16 & 26\end{bmatrix}$$

Matrix Multiplication

Definition

If the number of columns in a matrix A equals the number of rows in a matrix B, then product matrix AB exists.

Let A: $m \times n$ matrix, B: $n \times k$ matrix,

The **product matrix** $C = AB$ is a $m \times k$ matrix, with entries

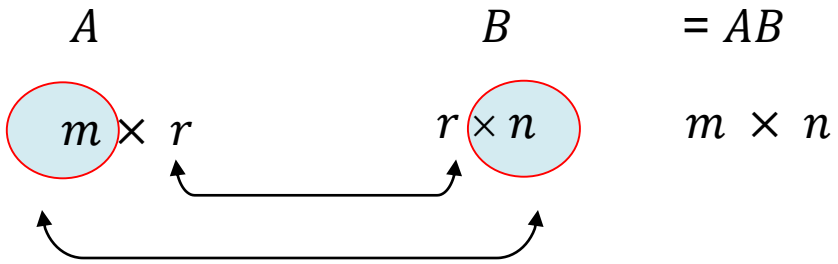
$$c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

- The entry c_{ij} is the dot product of the ith row of A, and the jth column of B: $c_{ij} = \vec{\boldsymbol{a}}_{\boldsymbol{i}}\boldsymbol{b}_j$
- Also, if $\boldsymbol{b}_j$ is the j-th row of B, then $AB = [A\boldsymbol{b_1}\ \ A\boldsymbol{b_2}\ \ A\boldsymbol{b_3}\ \ \ldots A\boldsymbol{b_k}]$.

If the number of columns in A does not equal the number of row B, **the product does not exist**.

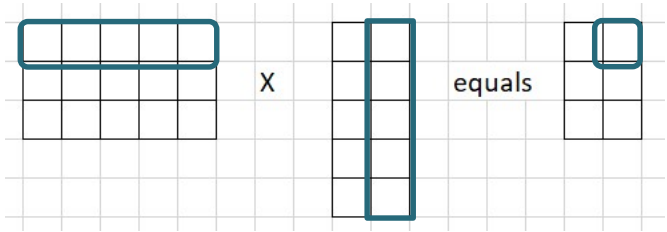
Size of a Product Matrix

If A is an $m \times r$ matrix and B is an $r \times n$ matrix, then AB will be an $m \times n$ matrix.



Example

If A is a 3×5 matrix and B is an 5×2 matrix.
A has five columns and B has five rows, thus AB exits, and AB *is* a 3×2 matrix.



Example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 5 & 0 & 1 \\ 3 & -2 & 6 \end{bmatrix}$$

Example

If $A = \begin{bmatrix} -1 & 2 & 5 \\ 0 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -1 & 4 \end{bmatrix}$, find AB and BA.

If $A = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 5 & -2 \\ -1 & 4 & 0 \end{bmatrix}$ and $\mathrm{x} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$, find Ax.

Solution

$$AB = \begin{bmatrix} -1 & 2 & 5 \\ 0 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} -6 & 13 \\ -3 & 16 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 2 & 5 \\ 0 & -2 & 3 \end{bmatrix} = \begin{bmatrix} -1 & -4 & 14 \\ 0 & 4 & -6 \\ 1 & -10 & 7 \end{bmatrix}$$

$$A\mathrm{x} = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 5 & -2 \\ -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ -9 \end{bmatrix}$$

Matrix multiplication and scalar multiplication

矩阵乘法不满足交换律!

- Scalar: ab=ba, commutative
- Matrix: AB might not equal BA

Example 1 矩阵乘法交换律不成立

Let $A = [1\ , 0]$ and $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$

Compute AB, BA, AC, CA

Solution $AB = 1; AC = 0;\ BA = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, CA=$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

Note. In general, $AB \neq BA$.

Example 2 矩阵乘法交换律不成立

Let $A=\begin{bmatrix}3 & 1 & 2\\ 4 & 1 & 5\end{bmatrix}, B=\begin{bmatrix}7 & 2\\ 6 & 3\end{bmatrix}$.

$$BA=\begin{bmatrix}7 & 2\\ 6 & 3\end{bmatrix}\begin{bmatrix}3 & 1 & 2\\ 4 & 1 & 5\end{bmatrix}$$

$$=\begin{bmatrix}\begin{bmatrix}7 & 2\end{bmatrix}\begin{bmatrix}3\\ 4\end{bmatrix} & \begin{bmatrix}7 & 2\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix} & \begin{bmatrix}7 & 2\end{bmatrix}\begin{bmatrix}2\\ 5\end{bmatrix}\\ \begin{bmatrix}6 & 3\end{bmatrix}\begin{bmatrix}3\\ 4\end{bmatrix} & \begin{bmatrix}6 & 3\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix} & \begin{bmatrix}6 & 3\end{bmatrix}\begin{bmatrix}2\\ 5\end{bmatrix}\end{bmatrix}=\begin{bmatrix}29 & 9 & 24\\ 30 & 9 & 27\end{bmatrix}$$

However if we try to calculate AB,

$$AB=\begin{bmatrix}3 & 1 & 2\\ 4 & 1 & 5\end{bmatrix}\begin{bmatrix}7 & 2\\ 6 & 3\end{bmatrix}$$

$$=\begin{bmatrix}\begin{bmatrix}3 & 1 & 2\end{bmatrix}\begin{bmatrix}7\\ 6\end{bmatrix} & \begin{bmatrix}3 & 1 & 2\end{bmatrix}\begin{bmatrix}2\\ 3\end{bmatrix}\\ \begin{bmatrix}4 & 1 & 5\end{bmatrix}\begin{bmatrix}7\\ 6\end{bmatrix} & \begin{bmatrix}4 & 1 & 5\end{bmatrix}\begin{bmatrix}2\\ 3\end{bmatrix}\end{bmatrix}$$

can not multiply, therefore AB does not exist.

Note. In general, $AB \neq BA$.

Algebraic Properties of Matrix Operations

Matrix addition and scalar multiplication

Theorem

Let A, B, and C be matrices and a, b, and c be scalars. Assume that the size of the matrices are such that the operations can be performed.

Properties of Matrix Addition and scalar Multiplication

1. $A + B = B + A$ — *Commutative property of addition* 加法交换律
2. $A + (B + C) = (A + B) + C$ — *Associative property of addition* 加法结合律
3. $A + O = O + A = A$ — (*where O is the appropriate zero matrix*)
4. $c(A + B) = cA + cB$ — *Distributive property of scalar multiplication*
5. $(a + b)C = aC + bC$ — *Distributive property of scalar multiplication*
6. $(ab)C = a(bC)$ — 标量乘法分配律

Algebraic Properties of Matrix Operations

Matrix multiplication

Theorem

Let A, B, and C be matrices and a, b, and c be scalars. Assume that the size of the matrices are such that the operations can be performed.

Properties of Matrix Multiplication

1. $A(BC) = (AB)C$	*Associative property of multiplication乘法结合律*
2. $A(B + C) = AB + AC$	*Distributive property of multiplication 乘法分配律*
3. $(A + B)C = AC + BC$	*Distributive property of multiplication 乘法分配律*
4. $AI_n = I_nA = A$	(*where I_n is the appropriate identity matrix*)
5. $c(AB) = (cA)B = A(cB)$	
Note: $AB \neq BA$ in general. *不满足交换律*	*Multiplication of matrices is not commutative.*

An $n \times n$ identity matrix is the matrix $I_n = (\delta_{ij})$, *where* $\delta_{ij} = \begin{cases} 1 & if\ i = j \\ 0 & if\ i \neq j \end{cases}$

Proof (*A*+*B*=*B*+*A*)

Consider the (*i*,*j*)th elements of matrices *A*+*B* and *B*+*A*:

$$(A+B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B+A)_{ij}.$$

$\therefore$ *A*+*B*=*B*+*A*

Proof A(BC)=(AB)C (optional, a bit messy)

Suppose the size of A, B, C are $m \times n$, $n \times p$ and $p \times q$.

Consider the (*i*,*j*)th elements of matrices A(*BC) and (AB)C*:

$$[A(BC)]_{ij} = \sum_{k=1}^{n} a_{ik}(BC)_{kj} = \sum_{k=1}^{n} a_{ik}\left(\sum_{t=1}^{p} b_{kt}c_{tj}\right)$$

Size of BC is $n \times q$

$$= \sum_{k=1}^{n}\sum_{t=1}^{p} a_{ik}b_{kt}c_{tj}$$

$$[(AB)C]_{ij} = \sum_{t=1}^{p} (AB)_{it}c_{tj} = \sum_{t=1}^{p} \left(\sum_{k=1}^{n} a_{ik}b_{kl}\right)c_{tj}$$

Size of AB is $m \times p$

$$= \sum_{t=1}^{p}\sum_{k=1}^{n} a_{ik}b_{kt}c_{tj} = \sum_{k=1}^{n}\sum_{t=1}^{p} a_{ik}b_{kt}c_{tj}$$

Example

Let $A=\begin{bmatrix}1 & 2\\ 3 & -1\end{bmatrix}$, $B=\begin{bmatrix}0 & 1 & 3\\ -1 & 0 & -2\end{bmatrix}$, and $C=\begin{bmatrix}4\\ -1\\ 0\end{bmatrix}$. Compute *ABC*.

Solution

A	B	C	=	D
2×2	2×3	3×1		2×1

$ABC = (AB)C = A(BC)$

$$AB=\begin{bmatrix}1 & 2\\ 3 & -1\end{bmatrix}\begin{bmatrix}0 & 1 & 3\\ -1 & 0 & -2\end{bmatrix}=\begin{bmatrix}-2 & 1 & -1\\ 1 & 3 & 11\end{bmatrix}.$$

$$(AB)C=\begin{bmatrix}-2 & 1 & -1\\ 1 & 3 & 11\end{bmatrix}\begin{bmatrix}4\\ -1\\ 0\end{bmatrix}=\begin{bmatrix}-9\\ 1\end{bmatrix}.$$

Exercise: first calculate (BC), then calculate A(BC), check if you get the same answer.

Caution

In algebra we know that the following **cancellation laws** apply.

- If $ab = ac$ and $a \neq 0$ then $b = c$.
- If $pq = 0$ then $p = 0$ or $q = 0$.

However the corresponding results are not true for matrices.

- $AB = AC$ does not imply that $B = C$.
- $PQ = \mathbf{0}$ does not imply that $P = \mathbf{0}$ or $Q = \mathbf{0}$.

Example

(1) Consider the matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 2 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} -3 & 8 \\ 3 & -2 \end{bmatrix}$.

Observe that $AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$, but $B \neq C$.

(2) Consider the matrices $P = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}$, and $Q = \begin{bmatrix} 2 & -6 \\ 1 & -3 \end{bmatrix}$.

Observe that $PQ = O$, but $P \neq O$ and $Q \neq O$.

Transpose of a Matrices

Definition

The **transpose** of a matrix A, denoted A^T, is the matrix whose columns are the rows of the given matrix A.

Thus $(A^T)_{ij} = A_{ji} \quad \forall i, j.$

Example

$$A = \begin{bmatrix} 2 & 7 \\ -8 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & -7 \\ 4 & 5 & 6 \end{bmatrix}, \text{and } C = \begin{bmatrix} -1 & 3 & 4 \end{bmatrix}.$$

$$A^T = \begin{bmatrix} 2 & -8 \\ 7 & 0 \end{bmatrix} \quad B^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ -7 & 6 \end{bmatrix} \quad C^T = \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}.$$

Theorem: Properties of Transpose

Let A and B be matrices and c be a scalar. Assume that the sizes of the matrices are such that the operations can be performed.

1. $(A + B)^T = A^T + B^T$ *Transpose of a sum*
2. $(cA)^T = cA^T$ *Transpose of a scalar multiple*
3. $(AB)^T = B^T A^T$ *Transpose of a product*
4. $(A^T)^T = A$

Example

If $A = \begin{bmatrix} 1 & 2 & -3 & 0 \\ 6 & 3 & 5 & -2 \end{bmatrix}$, find A^T.

If $A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$, verify that $(AB)^T = B^T A^T$.

(optional proofs)

Proof for 3. $(AB)^T = B^T A^T$

Suppose the size of A and B are $m \times n$ and $n \times p$.
Consider the (i,j)th elements of matrices $(AB)^T$ *and* $B^T A^T$:

$$(AB)^T_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

$$(B^T A^T)_{ij} = \sum_{k=1}^{n} (B^T)_{ik} (A^T)_{kj}$$
$$= \sum_{k=1}^{n} b_{ki} a_{jk}$$

Therefore, $(AB)^T = B^T A^T$

Symmetric Matrix

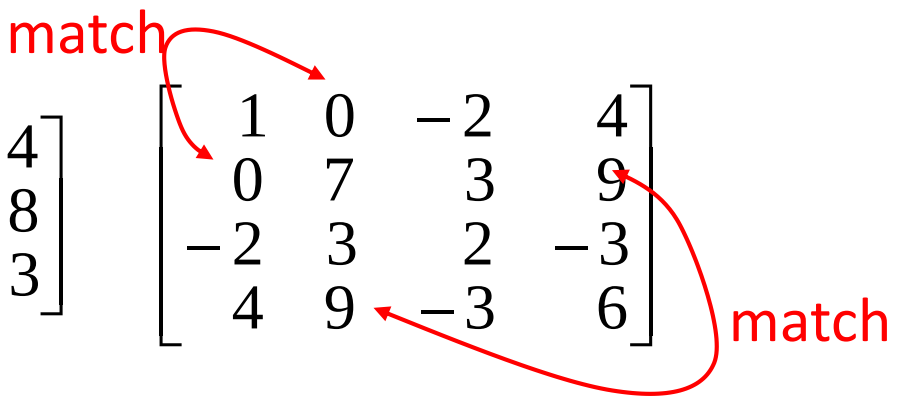
Definition

A **symmetric matrix** is a matrix that is equal to its transpose.

$$A = A^T, i.e., a_{ij} = a_{ji} \ \forall i, j$$

Example

$$\begin{bmatrix} 2 & 5 \\ 5 & -4 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & -4 \\ 1 & 7 & 8 \\ -4 & 8 & -3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 7 & 3 & 9 \\ -2 & 3 & 2 & -3 \\ 4 & 9 & -3 & 6 \end{bmatrix}$$



Example

Let A and B be symmetric matrices of the same size. Let C be a linear combination of A and B. Prove that C is symmetric.

Example

The matrix $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ has the property that $A^2 = O$

Is it possible for a nonzero symmetric 2×2 matrix to have this property?

No.
Why?

If $A = \begin{bmatrix} 1 & q & r \\ 5+r & -r & q-r \\ -3 & 5 & 6 \end{bmatrix}$, find q, r so that A is symmetric.

q=2 r=-3

If and only if

Let p and q be statements.
Suppose that p implies q (if p then q), written $p \Rightarrow q$,
and that also $q \Rightarrow p$, we say that

"p if and only if q" (iff) $\quad p \Leftrightarrow q$
(p is necessary and sufficient for q)

Example

Let A and B be symmetric matrices of the same size. Prove that the product AB is symmetric if and only if $AB = BA$.

Proof

*We have to show (a) AB is symmetric $\Rightarrow$ $AB = BA$, and the converse, (b) AB is symmetric $\Leftarrow$ $AB = BA$.

($\Rightarrow$) Let AB be symmetric, then

$AB = (AB)^T$ since AB is symmetric

$= B^T A^T$ by Thm (3)

$= BA$ since A and B are symmetric

Therefore, if AB is symmetric => AB=BA

($\Leftarrow$) Let $AB = BA$, then

$(AB)^T = (BA)^T$ since AB=BA

$= A^T B^T$ by Thm (3)

$= AB$ since A and B are symmetric

Therefore, if AB=BA => AB is symmetric.

Example

Let A be a symmetric matrix. Prove that A^2 is symmetric.

Proof

$$(A^2)^T = \quad (AA)^T = \quad (A^T A^T) = \quad AA = A^2$$

The Inverse of a Matrix

Definition

Let A be an $n \times n$ matrix. If a matrix B can be found such that $AB = BA = I_n$, then A is said to be **invertible** and B is called the **inverse** of A. If such a matrix B does not exist, then A has no inverse (denote $B = A^{-1}$, and $A^{-k} = (A^{-1})^k$).

Example

Prove that the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ has inverse $B = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$

Proof

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$BA = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Thus $AB = BA = I_2$, proving that the matrix *A* has inverse *B*.

Why we need inverse?

- Solve equation:

$$Ax = b$$
$$A^{-1}Ax = A^{-1}b$$
$$Ix = A^{-1}b$$
$$x = A^{-1}b$$

- Nonsingular = Invertible Matrices
  - A square $n \times n$ matrix A is ***nonsingular*** (invertible) if there exists a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$.
  - The matrix A^{-1}, if it exists, is called the inverse of A. If A does not have an inverse (A^{-1} does not exist), then A is ***singular*** (noninvertible).
  - Coefficient matrix is **nonsingular** = linear system is consistent, with **unique solution**.

Properties of Matrix Inverse

Let A and B be invertible matrices and c a nonzero scalar, Then

1. $(A^{-1})^{-1} = A$
2. $(cA)^{-1} = \frac{1}{c}A^{-1}$
3. $(AB)^{-1} = B^{-1}A^{-1}$
4. $(A^n)^{-1} = (A^{-1})^n$
5. $(A^T)^{-1} = (A^{-1})^T$

Proof

1. By definition, $AA^{-1} = A^{-1}A = I$.

2. $(cA)\left(\frac{1}{c}A^{-1}\right) = c\frac{1}{c}AA^{-1} = I = \left(\frac{1}{c}A^{-1}\right)(cA)$

3. $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I = (B^{-1}A^{-1})(AB)$

4. $A^n(A^{-1})^n = \underbrace{A \dots A}_{n} \cdot \underbrace{A^{-1} \dots A^{-1}}_{n} = \underbrace{A \dots A}_{(n-1)} \cdot I \cdot \underbrace{A^{-1} \dots A^{-1}}_{(n-1)} = \cdots = I = (A^{-1})^n A^n$

5. $AA^{-1} = I, (AA^{-1})^T = (A^{-1})^T A^T = I$

   $A^{-1}A = I, (A^{-1}A)^T = A^T(A^{-1})^T = I$

Theorem

- If A and B are both nonsingular $n \times n$ matrices, then the product AB is also nonsingular, and $(AB)^{-1} = B^{-1}A^{-1}$.

- If $A_1, \cdots, A_k$ are all nonsingular $n \times n$ matrices, then the product $A_1 A_2 \cdots A_k$ is nonsingular and

$$(A_1 A_2 \cdots A_k)^{-1} = {A_k}^{-1} \cdots {A_2}^{-1} {A_1}^{-1}$$

Example

If $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, verify that

$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$ and $(AB)^{-1} = B^{-1}A^{-1}$

Example

If $A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$, then it can be shown that A^{-1} $= \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$. Use this information to compute $(A^T)^{-1}$.

Solution

$$(A^T)^{-1} = (A^{-1})^T = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}.$$

Theorem

If a matrix has an inverse, then that inverse is unique.

Proof

Let B and C be inverses of A.

Thus $AB = BA = I_n$, and $AC = CA = I_n$.

Multiply both sides of the equation $AB = I_n$ by C.

$$C(AB) = CI_n$$
$$(CA)B = C$$
$$I_nB = C$$
$$B = C$$

Thus an invertible matrix has only one inverse.

Theorem

Let $AX = Y$ be a system of *n* linear equations in *n* variables.
If A^{-1} exists, the solution is unique and is given by $X = A^{-1}Y$.

Proof

($X = A^{-1}Y$ is a solution.)

Substitute $X = A^{-1}Y$ into the matrix equation.

$$AX = A(A^{-1}Y) = (AA^{-1})Y = I_nY = Y$$

(The solution is unique.)

Let X_1 be any solution, thus $AX_1 = Y$. Multiplying both sides of this equation by A^{-1} gives

$$A^{-1}AX_1 = A^{-1}Y$$
$$I_nX_1 = A^{-1}Y$$
$$X_1 = A^{-1}Y$$

As A^{-1} is unique, therefore the solution is unique.

Conclusions

- Addition, Scalar Multiplication
- Multiplication of Matrices
- Algebraic Properties of Matrix Operations
- Transpose, Symmetry, Inverse